Hearing Drones in 3D: Convolutional Detection and Direction-Finding from a Microphone Array

Shai Livshits, Dan Masad

Project Category: Applied – Audio / Convolutional Networks / Defense & Security

Problem Definition

Drones are hard to see and can fly with their radio link off, defeating cameras and radio detectors. Sound is a cheap fallback: rotors are loud and distinctive. From 24 ground microphones recording simultaneously, we want a network that reports ten times per second whether a drone is audible and in which direction – azimuth (compass bearing) and elevation (angle above the horizon).

Key Challenges

A distant drone is quiet and easily confused with wind, traffic, and aircraft. It fills a tiny fraction of the search space, so always answering "nothing" already scores well – severe class imbalance. Recordings are continuous, so neighbouring 100 ms windows look almost identical: a random split would leak training data into the test set.

Data

Western Sydney University's *UAV Acoustic Localization Dataset* (CC BY-SA 4.0, DOI 10.26183/07nm-4p35) downloads freely over HTTP, no registration: ~75 minutes of 24-channel outdoor recordings of a DJI Air 3 at two sites, ~10 minutes of drone-free background, and GPS logs every 100 ms giving true direction and speed – labels come for free. We downloaded a slice and confirmed audio, labels, and geometry agree. We train on one site and test on a second, unseen one.

Proposed Method

From sound to images: Sound from a given direction reaches the microphones at slightly different times. Delaying each microphone by exactly those offsets and adding the signals makes a correct guess reinforce and a wrong guess partly cancel. Sweeping a grid of 91 bearings $\times$ 23 elevations turns each 100 ms of audio into a 91×23 image whose pixels hold the energy arriving from each direction; the drone is a bright blob. This is fixed signal processing with nothing to learn, and it turns the task into an image problem.

Models: The baseline picks the brightest pixel, as classical systems do; it fails on noise and at range. Against it we train a small CNN outputting a per-pixel probability that the drone is there plus one overall "drone present" score, with class-weighted cross-entropy and Adam. Our main experiment feeds several consecutive images as extra input channels, letting the network use the fact that a real drone drifts smoothly while noise flickers. A secondary output regresses speed, testing whether the network picks up the Doppler pitch shift of an approaching drone.

Evaluation

On the unseen site: precision–recall curves and false alarms per hour (detection), median angular error and frames within 5/10/20° (direction), mean absolute error (speed). Ablations: number of stacked frames; accuracy with 12, 6, or 4 of the 24 microphones.

Related Work and Contribution

Rodríguez et al. (arXiv:2508.00307) published this dataset with a convolutional model that sees one frame at a time, named temporal context as future work, and released no training code. Al-Emadi et al. (2019) detect drones from one microphone, without direction. We contribute an open PyTorch implementation, the first use of consecutive frames on this dataset, a proper detection evaluation (the original reports only a five-minute false-alarm check), and a microphone-count study for cheaper arrays.

References

1. B. J. Rodríguez, S. F. Chevtchenko, M. Herrera Martínez, Y. Bethi, S. Afshar. *Acoustic Imaging for UAV Detection: Dense Beamformed Energy Maps and U-Net SELD.* arXiv:2508.00307. Dataset: DOI 10.26183/07nm-4p35.
2. S. Al-Emadi, A. Al-Ali, A. Mohammad, A. Al-Ali. *Audio Based Drone Detection and Identification using Deep Learning.* IWCMC, 2019.